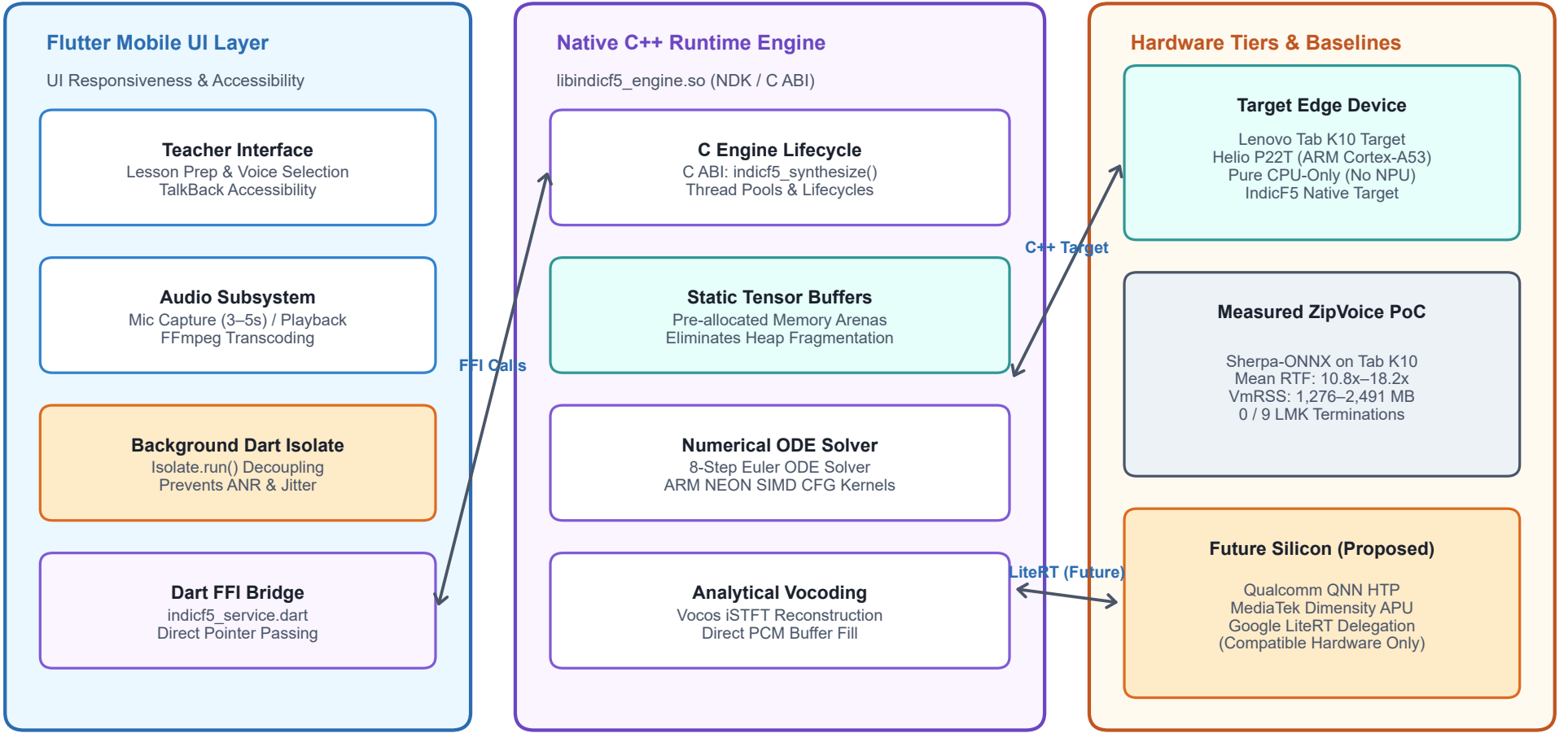

Flutter Mobile UI Layer
UI Responsiveness & Accessibility
Teacher Interface
Lesson Prep & Voice Selection
TalkBack Accessibility
Audio Subsystem
Mic Capture (3–5s) / Playback
FFmpeg Transcoding
Background Dart Isolate
Isolate.run() Decoupling
Prevents ANR & Jitter
Dart FFI Bridge
indicf5_service.dart
Direct Pointer Passing
FFI Calls
Native C++ Runtime Engine
libindicf5_engine.so (NDK / C ABI)
C Engine Lifecycle
C ABI: indicf5_synthesize()
Thread Pools & Lifecycles
Static Tensor Buffers
Pre-allocated Memory Arenas
Eliminates Heap Fragmentation
Numerical ODE Solver
8-Step Euler ODE Solver
ARM NEON SIMD CFG Kernels
Analytical Vocoding
Vocos iSTFT Reconstruction
Direct PCM Buffer Fill
C++ Target
LiteRT (Future)
Hardware Tiers & Baselines
Target Edge Device
Lenovo Tab K10 Target
Helio P22T (ARM Cortex-A53)
Pure CPU-Only (No NPU)
IndicF5 Native Target
Measured ZipVoice PoC
Sherpa-ONNX on Tab K10
Mean RTF: 10.8x–18.2x
VmRSS: 1,276–2,491 MB
0 / 9 LMK Terminations
Future Silicon (Proposed)
Qualcomm QNN HTP
MediaTek Dimensity APU
Google LiteRT Delegation
(Compatible Hardware Only)